Minimum Window Substring (hard)

Problem Statement

Given a string `s` and `t` of length `m` and `n` respectively, find the `smallest substring` in a given string 's' that contains all the characters (including duplicate) of another string 't'. The `order of characters` in 't' doesn't matter, but the substring in 's' must include all characters of 't', possibly with additional characters.

If no such substring exists, return an `empty` string.

Examples

Example 1:

- **Input**: s = "xyyzyzyx", t = "xyz"
- **Expected Output**: "zyx"
- **Justification**: "zyx" is the shortest substring of 's' that contains 'x', 'y', and 'z'.

Example 2:

- **Input**: s = "abracadabra", t = "aa"
- **Expected Output**: "ada"
- **Justification**: "ada" at the end of 's' is the shortest substring containing all 'a's in 't'.

Example 3:

- **Input**: s = "welcomeback", t = "ole"
- **Expected Output**: "elco"
- **Justification**: "elco" includes 'o', 'l', and 'e', fulfilling the requirements with the minimum length.

Try it yourself

Try solving this question here:

